A Within-Layer Adjacency Lemma

A small-ball first-moment estimate for disjoint adjacencies among subset sums

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Abstract

We prove a first-moment small-ball estimate for subset sums of i.i.d. uniform variables. The estimate is the within-layer input to a golden-ratio lower bound for Erdős problem #1100. Let $x_1, \dots, x_k$ be i.i.d. Uniform$[0,1]$, let $X(S) = \sum_{i \in S} x_i$, and for $j = \lfloor \beta k \rfloor$ order the $\binom{k}{j}$ size-j sums on the line; let $N(k,j)$ be the expected number of *consecutive disjoint* pairs — size-j sets S, T with $S \cap T = \varnothing$ and no size-j sum strictly between $X(S)$ and $X(T)$. We show $N(k,j) \gg \binom{k-j}{j}$ for β near 0.276. Summed over layers via $\sum_j \binom{k-j}{j} = F_{k+1}$, it yields the base φ in that bound, $g_{\mathrm{sf}}(k) \gg \varphi^k$ for coprime adjacent divisors of squarefree integers, developed in companion notes. The proof is a truncated first moment whose only analytic inputs are elementary: standard Irwin–Hall density facts and a Hoeffding tail bound, with no local limit theorem or Edgeworth expansion. Its one delicate point is the overlap-free "swap" blocker class, controlled by a codimension-two small-ball bound — blocking forces two *independent* signed forms into a window of width θ, costing θ^2 rather than θ, which makes that class exponentially negligible.

1 Statement

The probability model. Let $x_1, \dots, x_k$ be i.i.d. Uniform$[0,1]$, with mean $\frac{1}{2}$ and variance $s^2 = \frac{1}{12}$, and for $S \subseteq [k]$ write $X(S) = \sum_{i \in S} x_i$. Fix $\beta \in (0, \frac{1}{2})$ and $j = \lfloor \beta k \rfloor$, so $2j \le k$ for large k. Order the $\binom{k}{j}$ values $\{X(S) : |S| = j\}$ increasingly. Two size-j subsets S, T are *consecutive* if no size-j subset R has $X(R)$ strictly between $X(S)$ and $X(T)$, and *disjoint* if $S \cap T = \varnothing$. Set

$$N(k,j) = \mathbb{E}\big[\, \#\{\{S,T\} : \ |S| = |T| = j, \ S \cap T = \varnothing, \ S,T \text{ consecutive}\}\,\big].$$

Lemma 1. *For each $\beta \in (0, \frac{1}{2})$ there is $c_\beta > 0$ such that, with $j = \lfloor \beta k \rfloor$,*

$$N(k,j) \ \ge \ c_\beta \binom{k-j}{j} \qquad (k \to \infty).$$

This is the entire content of the note; the statement above is self-contained and uses nothing beyond the uniform model just defined.

Context. In a tight-band (lexicographic) limit the coprime adjacent divisor count of a squarefree $n = \prod_{i \le k} p_i$ decomposes over cardinality layers as $\sum_j N(k,j) + O(k)$; with $\sum_j \binom{k-j}{j} = F_{k+1} \asymp \varphi^k$ and $\varphi = (1+\sqrt{5})/2$, Lemma 1 yields $g_{\mathrm{sf}}(k) \gg \varphi^k$, improving the Erdős–Simonovits lower bound $(\sqrt{2} + o(1))^k$ for Erdős problem #1100. The layer reduction and the transfer from uniform weights to prime weights $a_i = \log p_i$ are carried out in companion notes [2, 3]; the transfer is the one step of the larger program not contained here. Nothing below depends on either. The companion bound rests entirely on the lemma proved here; its proof is, to the author's knowledge, complete as written but has not been independently refereed, which makes the uniformity of c_β across the saddle the natural point for outside scrutiny.

What the proof uses. Only one-dimensional facts about Irwin–Hall densities (the law of a sum of independent Uniform$[0,1]$ variables): boundedness, C^1-smoothness, and the L^2 norm. The small-ball event $\{|\Delta| \le \theta\}$ enters only through the *density* of the gap Δ at exponentially small scales, never through its distribution function. A CDF tool such as Berry–Esseen, with additive error $\asymp j^{-1/2}$, is therefore both too weak and unnecessary: the exact identity $f_\Delta(0) = \|f_{\mathrm{IH}_j}\|_2^2$ supplies the value directly, and C^1-smoothness transports it across the narrow window. The only scale that matters is the polynomial one, $j^{-1/2}$ relative to the spread of Δ; no 2^{-k}-scale anti-concentration enters.

2 Three ingredients

Proposition 1 (Irwin–Hall density bounds)**.** *There is a constant $A < \infty$, uniform in k, such that for disjoint S, T and for R with $|S| = |T| = |R| = j$, $R \notin \{S, T\}$, writing $m = |R \setminus (S \cup T)|$:*

(i) The gap $\Delta = X(T) - X(S)$ has a C^1 density f_Δ with

$$f_\Delta(0) = \|f_{\mathrm{IH}_j}\|_2^2 \;\ge\; \frac{c_0}{\sqrt{j}} \quad (c_0 > 0),$$

f_{IH_j} the order-j Irwin–Hall density. Hence, since $f_\Delta \in C^1$ and $\theta \to 0$, $\mathbb{P}[\,|\Delta| \le \theta\,] = 2\theta f_\Delta(0)(1 + o(1))$ uniformly in j.

(ii) The conditional density of $X(R)$ given $(X(S), X(T))$ satisfies

$$\sup_w f_{X(R)|X(S),X(T)}(w) \;\le\; \frac{A}{s\sqrt{\max(m,1)}}.$$

Proof. (i) With $A_0 = \sum_{i\in S} x_i$ and $B_0 = \sum_{i\in T} x_i$ independent, each Irwin–Hall of order j, $\Delta = B_0 - A_0$ has density $f_\Delta(t) = \int f_{\mathrm{IH}_j}(u) f_{\mathrm{IH}_j}(u+t)\,du$, so $f_\Delta(0) = \|f_{\mathrm{IH}_j}\|_2^2$. By Chebyshev at least half the Irwin–Hall mass lies in an interval I of length $O(\sqrt{j})$ about $j/2$; Cauchy–Schwarz gives $\frac{1}{2} \le \int_I f_{\mathrm{IH}_j} \le \|f_{\mathrm{IH}_j}\|_2 |I|^{1/2}$, whence $f_\Delta(0) \ge c_0/\sqrt{j}$. For $j \ge 2$, $f_\Delta \in C^1$ with $\|f_\Delta'\|_\infty = O(1)$, so on the exponentially small window $|t| \le \theta$ the variation is $|f_\Delta(t) - f_\Delta(0)| \le \|f_\Delta'\|_\infty \theta = o(f_\Delta(0))$, giving the asymptotic uniformly in j. (ii) Conditioning on $X(S), X(T)$ leaves the m coordinates of $R \setminus (S \cup T)$ free, so the conditional law of $X(R)$ is a convolution with the order-m Irwin–Hall law; convolution does not increase the supremum, so $\sup_w f_{X(R)|X(S),X(T)}(w) \le \|f_{\mathrm{IH}_m}\|_\infty \le A/(s\sqrt{m})$ for $m \ge 1$. For $m = 0$ there is no free coordinate and (ii) does not apply; those terms are carried by Proposition 3. □

Proposition 2 (External blockers concentrate)**.** *Let B_{ext} count blockers R ($|R| = j$, $R \ne S_0, T_0$, $X(R)$ in the gap) with defect $m = |R \setminus (S_0 \cup T_0)| \ge 1$. Then for every $\theta > 0$,*

$$\mathbb{E}\big[B_{\mathrm{ext}} \mathbf{1}\{|\Delta| \le \theta\}\big] \;\le\; \frac{A\theta}{s}\, \mathbb{P}[\,|\Delta| \le \theta\,]\, \Phi, \qquad \Phi = \sum_{m \ge 1} \frac{\binom{2j}{j-m}\binom{k-2j}{m}}{\sqrt{m}},$$

and $\Phi \asymp \binom{k}{j}/\sqrt{j}$; precisely $c_1 \binom{k}{j}/\sqrt{j} \le \Phi \le C_2 \binom{k}{j}/\sqrt{j}$ with $0 < c_1 \le C_2 = C_2(\beta) < \infty$.

Proof. For a single R with $m \ge 1$, condition on $(X(S_0), X(T_0))$ and use Proposition 1(ii) with $|\mathrm{gap}| = |\Delta| \le \theta$: $\mathbb{P}[X(R) \in \mathrm{gap} \mid X(S_0), X(T_0)] \le A\theta/(s\sqrt{m})$. The number of size-$j$ subsets with defect m is $\binom{2j}{j-m}\binom{k-2j}{m}$ (choose $j - m$ inside the fixed $2j$-set $S_0 \cup T_0$ and m outside); summing and taking expectations gives the displayed bound with Φ as stated. The $m = 0$ class is excluded — Proposition 1(ii) does not reach it — and is carried by B_{int}.

For the size of Φ: by Vandermonde $\sum_{m\ge 0} \binom{2j}{j-m}\binom{k-2j}{m} = \binom{k}{j}$, so the weights $\binom{2j}{j-m}\binom{k-2j}{m}/\binom{k}{j}$ are the law of $m \sim \mathrm{Hypergeometric}(k, k-2j, j)$, the number of elements of a uniform random

j-subset outside $S_0 \cup T_0$, with mean $\bar{m} = (1-2\beta)j + O(1)$. Thus $\Phi = \binom{k}{j}\,\mathbb{E}[m^{-1/2}\mathbf{1}\{m \geq 1\}]$. Split at $\bar{m}/2$: on $\{m \geq \bar{m}/2\}$, $m^{-1/2} \leq \sqrt{2}\,\bar{m}^{-1/2}$; on $\{1 \leq m < \bar{m}/2\}$, $m^{-1/2} \leq 1$ and Hoeffding for the hypergeometric law (same one-sided bounds as the binomial of equal mean) gives $\mathbb{P}[m \leq \bar{m}/2] \leq \exp(-\frac{1}{2}(1-2\beta)^2 j + o(j))$. Hence $\mathbb{E}[m^{-1/2}\mathbf{1}\{m \geq 1\}] \leq \sqrt{2}\,\bar{m}^{-1/2} + e^{-cj} \leq C_2/\sqrt{j}$. The matching lower bound $\mathbb{E}[m^{-1/2}\mathbf{1}\{m \geq 1\}] \geq (2\bar{m})^{-1/2}\,\mathbb{P}[m \geq \bar{m}/2] \geq c_1/\sqrt{j}$ follows from the same concentration. This gives the two-sided bound on Φ. □

Proposition 3 (Internal blockers: a codimension-two bound)**.** *Fix disjoint S_0, T_0 with $|S_0| = |T_0| = j$, put $\Delta = X(T_0) - X(S_0)$, and let*

$$B_{\text{int}} = \#\big\{R \subseteq S_0 \cup T_0 :\ |R| = j,\ R \neq S_0, T_0,\ X(R) \in (X(S_0) \wedge X(T_0),\, X(S_0) \vee X(T_0))\big\}$$

count the overlap-free ($m = 0$) blockers. Then for every $\theta > 0$,

$$\mathbb{E}\big[B_{\text{int}}\mathbf{1}\{|\Delta| \leq \theta\}\big] \ \leq\ \frac{2A^2}{\sqrt{j-1}}\binom{2j}{j}\theta^2.$$

Remark 1 (Why the internal case needs its own bound)**.** *An internal blocker lies inside $S_0 \cup T_0$ and so carries* no *coordinate outside the pair; the conditional-density bound of Proposition 1(ii), which conditions on a free coordinate, does not apply to $X(R)$ directly. The proof below substitutes a codimension-two mechanism: the swap decomposition writes the blocking event as the joint localization of* two independent *signed forms U and V, to each of which Proposition 1(ii) applies separately, so the event costs θ^2 rather than θ. This is the one point in the argument where the generic blocker estimate fails and is replaced rather than reused.*

Proof. Take $\Delta > 0$ (the case $\Delta < 0$ is symmetric, supplying the factor 2). A set $R \subseteq S_0 \cup T_0$ with $|R| = j$ is determined by $A = S_0 \setminus R \subseteq S_0$ and $B = R \cap T_0 \subseteq T_0$ with $|A| = |B| = r$; then $R = (S_0 \setminus A) \cup B$, and $R \neq S_0, T_0$ forces $1 \leq r \leq j-1$. Put

$$U = X(B) - X(A) \text{ (coordinates } A \cup B), \qquad V = X(T_0 \setminus B) - X(S_0 \setminus A) \text{ (coordinates } (S_0 \setminus A) \cup (T_0 \setminus B)).$$

The coordinate sets are disjoint, so $U \perp V$; moreover $\Delta = U + V$ and $X(R) - X(S_0) = U$. Hence "R blocks and $0 < \Delta \leq \theta$" is exactly $\{U > 0, V > 0, U + V \leq \theta\}$, forcing $U \in (0, \theta]$ *and* $V \in (0, \theta]$. By independence and Proposition 1(ii) applied to the r- and $(j-r)$-fold forms,

$$\mathbb{P}[R \text{ blocks},\, 0 < \Delta \leq \theta] \leq \theta \|f_U\|_\infty \cdot \theta \|f_V\|_\infty \leq \frac{A^2\theta^2}{\sqrt{r(j-r)}}.$$

This is the codimension-two gain: one factor θ from the gap, a second from the independent swap form. Summing over the $\binom{j}{r}^2$ choices of (A, B) with r swaps, using $\sqrt{r(j-r)} \geq \sqrt{j-1}$ and Vandermonde $\sum_{r=0}^{j}\binom{j}{r}^2 = \binom{2j}{j}$,

$$\sum_R \mathbb{P}[\cdots] \leq \frac{A^2\theta^2}{\sqrt{j-1}}\sum_{r=1}^{j-1}\binom{j}{r}^2 \leq \frac{A^2\theta^2}{\sqrt{j-1}}\binom{2j}{j},$$

and the factor 2 for $\Delta < 0$ gives the claim. □

3 Proof of Lemma 1

Reduction to a single pair. All unordered disjoint pairs are exchangeable, so with $D_k = \frac{1}{2}\binom{k}{j}\binom{k-j}{j}$ their number,

$$N(k, j) = D_k\, P_{\text{cons}}, \qquad P_{\text{cons}} = \mathbb{P}[\, S_0, T_0 \text{ consecutive}\,]$$

for one fixed disjoint pair (S_0, T_0). It suffices to show $P_{\text{cons}} \geq c'_\beta/\binom{k}{j}$.

Truncated first moment. Let B count all blockers of $\{S_0, T_0\}$, so $\{\text{consecutive}\} = \{B = 0\}$. On $\{|\Delta| \le \theta\}$ we have $\mathbf{1}\{B = 0\} \ge 1 - B$, hence

$$P_{\text{cons}} = \mathbb{E}\,\mathbf{1}\{B = 0\} \;\ge\; \mathbb{P}[\,|\Delta| \le \theta\,] - \mathbb{E}\big[B\mathbf{1}\{|\Delta| \le \theta\}\big]. \tag{1}$$

Split $B = B_{\text{ext}} + B_{\text{int}}$ by defect $m \ge 1$ resp. $m = 0$. Propositions 2 and 3 bound the two pieces.

Window and assembly. Set $\theta = s/(2A\Phi)$, so Proposition 2 gives $\mathbb{E}[B_{\text{ext}}\mathbf{1}\{|\Delta| \le \theta\}] \le \frac{1}{2}\,\mathbb{P}[\,|\Delta| \le \theta\,]$, and the two-sided bound on Φ pins the window,

$$\frac{s\sqrt{j}}{2AC_2\binom{k}{j}} \;\le\; \theta \;\le\; \frac{s\sqrt{j}}{2Ac_1\binom{k}{j}} \;\asymp\; \frac{\sqrt{j}}{\binom{k}{j}}.$$

For the internal piece, Proposition 3 with $\mathbb{P}[\,|\Delta| \le \theta\,] \ge 2c_0\theta/\sqrt{j}\,(1 + o(1))$ (Proposition 1(i)) gives, writing H for the binary entropy,

$$\frac{\mathbb{E}[B_{\text{int}}\mathbf{1}\{|\Delta| \le \theta\}]}{\mathbb{P}[\,|\Delta| \le \theta\,]} \;\asymp\; \frac{A^2\binom{2j}{j}}{c_0}\theta \;\le\; \frac{As}{2c_0c_1}\,\frac{\sqrt{j}\,\binom{2j}{j}}{\binom{k}{j}} \;=\; O\big(\sqrt{j}\,2^{(2\beta - H(\beta))k}\big) \;\longrightarrow\; 0,$$

since $\binom{2j}{j} \le 2^{2\beta k}$, $\binom{k}{j} = 2^{(H(\beta)+o(1))k}$, and $H(\beta) > 2\beta$ for all $\beta < \frac{1}{2}$ (e.g. $H(0.276) = 0.850 > 0.552$). The internal blockers are thus an exponentially small fraction of the main term. Combining,

$$\mathbb{E}\big[B\mathbf{1}\{|\Delta| \le \theta\}\big] \le \big(\tfrac{1}{2} + o(1)\big)\,\mathbb{P}[\,|\Delta| \le \theta\,],$$

so (1) gives $P_{\text{cons}} \ge (\frac{1}{2} - o(1))\,\mathbb{P}[\,|\Delta| \le \theta\,]$. By Proposition 1(i), $\mathbb{P}[\,|\Delta| \le \theta\,] \ge 2c_0\theta/\sqrt{j}\,(1 + o(1))$ and $\theta \ge s\sqrt{j}/(2AC_2\binom{k}{j})$, so

$$P_{\text{cons}} \;\ge\; \frac{c_0\theta}{\sqrt{j}}(1 + o(1)) \;\ge\; \frac{c_0 s}{2AC_2} \cdot \frac{1 + o(1)}{\binom{k}{j}}.$$

Finally $N(k, j) = D_k P_{\text{cons}} \ge \dfrac{c_0 s}{4AC_2}\binom{k-j}{j}(1 + o(1)) = c_\beta\binom{k-j}{j}$, with $c_\beta = c_0 s/(4AC_2(\beta)) > 0$. □

4 Remarks

Remark 2 (Why isoperimetry does not shortcut ingredient (i))**.** *It is tempting to replace Proposition 1(i) by a Harper/Erdős–Littlewood–Offord crowding argument, as in the distinct-subset-sums problem [1, 5], where a vertex-isoperimetric bound forces many signed sums into a short interval and pigeonhole extracts a close pair. That route is structurally unavailable here: the close pairs it produces differ in a single coordinate (Hamming distance* 1*), whereas the pairs counted by* $N(k, j)$ *are* disjoint *(Hamming distance* $2j$*). Isoperimetry crowds Hamming-adjacent points; the present count needs crowding among Hamming-opposite points, and no slicing reconciles the two. The shared content with that literature is the small-ball density toolkit (Proposition 1), not the combinatorial mechanism.*

Remark 3 (Constants and the role of φ)**.** *A and c_β are not optimized; the robust output is the* exponent*, which depends only on the shape* $N(k, j) \asymp \binom{k-j}{j}$*, not on the* $O(1)$ *prefactors. In the larger program φ is a floor rather than the growth rate: finite-k maxima exceed the φ^k lower bound by bounded factors, consistent with $g_{\text{sf}}(k) \asymp \varphi^k$ but not establishing a larger base, so whether the true rate equals φ or exceeds it is left open, between φ and the unconditional upper bound $g_{\text{sf}}(k) \le (1.88988\ldots)^k$ proved in the companion note [4].*

References

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